



Leaf chlorophyll plasticity in advanced regeneration of co-occurring *Acer pensylvanicum* and *Fagus grandifolia* beneath deciduous forest canopy

Olaf Thomas Bouman

Department of Biology, Cape Breton University, Sydney, Nova Scotia, B1P6L2, Canada

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ABSTRACT

The chemical trait 'total chlorophyll content per unit leaf area' (CHL) has long been recognized to confer plasticity within a genetically fixed range of phenotypic variance. The plasticity of leaf CHL (PCHL) was assessed with special reference to saplings of the two morphologically divergent tree species *Acer pensylvanicum* L. (AP) and *Fagus grandifolia* Ehrh. (FG) which co-occur in the eastern deciduous forest biome of North America. A randomized block design with repeated measures was used to field-test the hypothesis that low leaf display efficiency favoured selection for higher PCHL in FG than in AP. Species were tested in the shade of closed forest canopy at naturally low soil fertility and following applications of mineral NPK to test the second hypothesis that nutrient-poor soils select for high PCHL. Results support the first hypothesis of higher phenotypic PCHL and higher developmental PCHL in FG than in AP. Moreover, a markedly more dynamic CHL increase coincided with more rapid leaf development in FG compared to AP. Differential leaf phenology likely contributed to high PCHL in FG. Results also support, in the case of FG, the second hypothesis that nutrient-poor soils favour phenotypic PCHL. This is, however, not true for AP and confirms a lower shade tolerance rank for AP. Results strongly suggest that the ecological conditions for advanced regeneration in deep shade on nutrient-poor soils maintain high PCHL in FG populations.

1. Introduction

The chemical trait 'total chlorophyll content per unit leaf area' (CHL) represents an adaptive trait of light harvesting systems that correlates with photosynthetic rates of leaves and the productivity of ecosystems (Evans, 1989; Yang et al., 2017). The trait has long been recognized to confer capacity to adjust to varying growing environments, often referred to as plasticity within a genetically fixed range of phenotypic variance (i.e., reaction norm; Detmer, 1918).

As the key chemical for carbon (C) assimilation, knowledge of the plasticity of chlorophyll (PCHL) is integral to understanding the organismal capacity of tree species to cope with varying light exposure and mineral resource limitations. Numerous researchers have used mechanistic approaches that combine morphometric and physiological plant traits to rank the capacity of tree species to tolerate shade. However, Loach (1970) stated already decades ago that no clear-cut distinction could be drawn between tolerant and intolerant species on the basis of the relative size or efficiency of their photosynthetic systems in low light (i.e., deep shade). More recently, Körner (2005) reiterated this assertion and Valladares and Niinemets (2008) acknowledged the importance of trait plasticity for shade tolerance. In fact, Valladares et al. (2002) linked high shade tolerance at the seedling

stage of *Fagus sylvatica* to PCHL. For saplings of the same species, Bouman (2014) reported a size-dependent increase of PCHL in advanced regeneration suggesting optimization of limited mineral N utilization that is essential for chlorophyll function within the expanding leaf area of the sapling crown (cf. Körner, 2008). Thus, PCHL is an expression of organismal homeostasis. Light exposure of individual leaves is however, not only limited by competing vegetation such as forest canopy but also by other leaves on the same plant hence self-shading. Morphometric analysis of shoot systems and foliage suggests inherent differences in self-shading among co-occurring tree species (Lei and Lechowicz, 1990; Beaudet and Messier, 1998). Moreover, the importance of leaf C assimilation for plant growth rate increases relative to mineral nutrient supply in deep shade (Körner, 2008) although the growth response to nutrient amendments at low light exposure is species-dependent among northern hardwood species (Canham et al., 1996; Bouman et al., 2012). The relationship between PCHL and self-shading as well as the relationship between PCHL and mineral nutrition are not well known for most tree species. This knowledge gap relates directly to the broad research question that emanated from a comprehensive review of an exponentially increasing number of scientific publications about plasticity as a biological phenomenon, namely as to what ecological conditions maintain plasticity (Forsman, 2015).

E-mail address: thomas.bouman@cbu.ca.

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The tree species *Acer pensylvanicum* L. (AP) of the Sapindales is known for its low wood density, low wood N content, sparse foliage with opposite arrangement of large leaves, and a high ratio of leaf area over shoot length thus high leaf display efficiency (Lei and Lechowicz, 1990; King, 2003; Martin et al., 2015). The species is indicative of old-growth broadleaved forest types (Whitney and Foster, 1988). Sharp increases of relative growth rate along light gradients from understory light conditions to forest gaps have been observed in AP (Sipe and Bazzaz, 1995; Lei and Lechowicz, 1998). However, the short-lived AP rarely grows to tree size albeit high AP abundance at the seedling stage (D'Amato et al., 2015). In contrast, high resource investments in a durable, highly ramified, densely foliated shoot system (with medium-size leaves at low leaf display efficiency), and limited response of growth rate to increased light exposure characterize the long-lived *Fagus grandifolia* Ehrh. (FG) of the Fagales (Beaudet and Messier, 1998; Martin et al., 2015). The two morphologically divergent species interperse naturally within small areas at low light exposure beneath mature forest canopy in the eastern deciduous forest biome of North America (D'Amato et al., 2015). Jurik (1986) reported for FG a slightly lower value for the morphometric Leaf Mass per Area (LMA) than for co-occurring AP in an exploratory field study in Michigan. Albeit the significance of LMA for shade tolerance in mechanistic growth models, Aber and Melillo (2001) questioned the ranking of AP as a shade-tolerant species based on a literature review with a geographic focus on the native range of AP. Subsequently, Niinemets and Valladares (2006) assigned on a 5-point scale a lower shade tolerance rank of 3.56 to AP compared to 4.75 for FG based on a literature review for a large number of tree species distributed across continents.

This study optically assessed CHL repeatedly on the same leaf in four different crown positions during the 2015 season and in the same four crown positions in 2016 on co-occurring AP and FG at the sapling-stage beneath mature deciduous forest canopy. The experimental design allowed for testing the hypothesis that low leaf display efficiency favoured selection for higher PCHL in FG than in AP due to increased self-shading at low leaf display efficiency. Each species was tested at naturally low soil fertility and following applications of mineral NPK to test the hypothesis that nutrient-poor soils select for high PCHL due to the essential role of mineral N and P for chlorophyll-associated protein synthesis. For the purpose of this paper, phenotypic PCHL (PPCHL) refers to PCHL assessed for a single datum and developmental PCHL (DPCHL) refers to PCHL based on repeated measurements during the growing season (Ehrendorfer, 1998; Forsman, 2015).

2. Materials and methods

2.1. Research location

In the Province of Nova Scotia on the Atlantic coast of Canada, naturally growing saplings of AP and FG were tested in a deciduous hardwood near the rural community of East Bay (46°N; 60° 01' W; 100 m a.s.l.). Compared to long-term averages, the forest on coarse-textured soil experienced wetter than normal weather conditions in 2015 and drier than normal conditions in 2016 (Fig. 1; <https://weather.gc.ca>). Seasonal rainfall amounted to 356 mm during the period 16 June to 19 September 2015 compared to 202 mm from 15 June to 11 September 2016. Growing degree days above 5 °C was 1140 in 2015 and 1107 in 2016. Forest history, shade effects on sapling survival of other woody species, and ecosystem conditions in the East Bay Community Forest have previously been characterized by the author (Bouman et al., 2004, 2012; Bouman, 2018). Dominant tree species in approximately 16-m-high forest canopy comprised *A. rubrum*, *A. saccharum*, and *Betula alleghaniensis* with minor admixture of FG. Acid soil and tight nutrient cycling characterize local and regional ecosystem conditions (Bouman, 2018). At the landscape level, experimental plots are embedded within a large patch of deciduous forest that does not have a history of clearing for agriculture (Bouman et al., 2004).

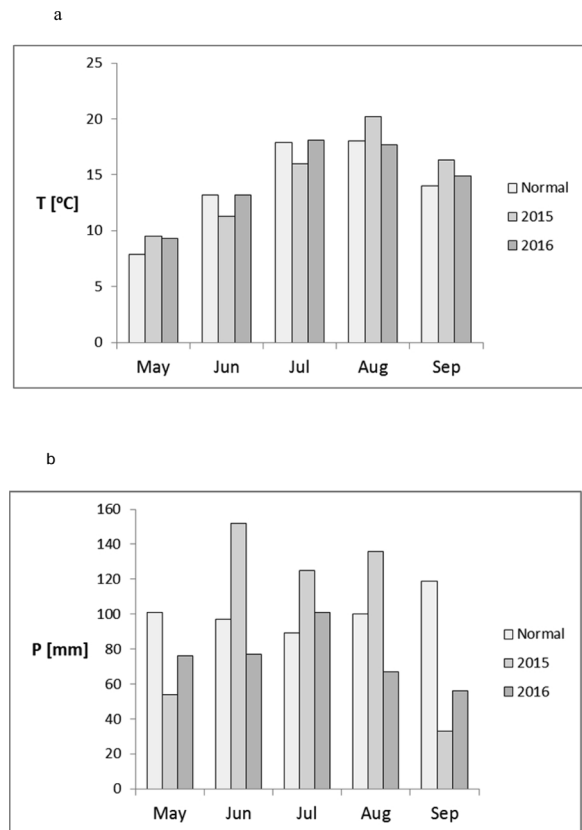


Fig. 1. Mean monthly temperature (T; 1a) and monthly precipitation (P; 1b) for nearest meteorological station to East Bay Community Forest; <https://weather.gc.ca>: "Normal" is for the period 1980–2010.

2.2. Experimental design

The experiment was conducted in six areas of 0.16 ha each with advanced regeneration of AP and FG beneath closed deciduous canopy within the same deciduous forest of approximately 50 ha. In each area, four saplings of AP with a stem length greater than 0.6 m were randomly selected and then the nearest FG sapling, with comparable stem length, for each AP sapling. Two species pairs received NPK within a 0.3-m radius around the sapling stem. Thus, the experiment comprised four treatments: AP at natural soil fertility (AP-CON), AP with application of NPK (AP-NPK), FG-CON, and FG-NPK. Commercially available NPK fertilizer with a 10:5:5-mixture was applied in split applications following snow melt at a rate of 50 kg N ha⁻¹ in late May, June, and July 2015; and, late April and mid June 2016. Thus, fertilized saplings received a total of 250 kg N ha⁻¹. Sapling crowns were standing freely not encumbered by adjacent saplings. Each sapling was labelled using a combination of flag tape for visibility and aluminum plates for durability. Aluminum rings were used to mark the leading shoot and a mid-level lateral shoot on each sapling and a proximal and distal petiole on each shoot for repeated measurement of CHL on the same leaf during the 2015 growing season and on the same twig in 2016.

2.3. Sampling and data collection

A hemispherical densitometer was used to measure forest canopy closure above saplings during full flush in July and in late season September 2015. For non-destructive monitoring of CHL, readings of an optical index were taken *in vivo* using the CCM-200 plus v1.00 device (OPTI-SCIENCES Inc., Hudson, NH; Van der Berg and Perkins, 2004) around noon between 11 h and 13 h on 30 May, 16 June, 8 July, 4 August, and 19 September of 2015 and 15 June and 11 September of

2016. In June 2015, measurements of stem length, total shoot length (i.e., stem length plus sum of all branch lengths), and leaf counts were taken to characterize above-ground sapling structure. Leaf size development was characterized by leaf length measured 30 May, 16 June, and 4 August. In August 2015, one leaf was clipped in each of the four crown positions. The fresh turgid leaves were photocopied prior to drying in a ventilated cabinet at 70 °C for three days for gravimetric determination of leaf area (Hendry and Grime, 1993) and relative water content (RWC) using a Sartorius (Göttingen, Germany) precision balance. Optical chlorophyll index readings were taken immediately in the forest on the freshly clipped leaves. Single leaf blade mass was determined following the separation of the blade from the petiole. The four-leaf sample collected from each sapling was subsequently pooled and submitted to an analytical lab on Dalhousie University Agricultural Campus for determination of total leaf blade tissue nitrogen using a dry combustion LECO analyzer.

2.4. Calculations and data analysis

The total sapling leaf area was calculated by multiplying the area sum of the 4-leaf sample by the total leaf count divided by 4. Shoot morphology was characterized by dividing the stem-length by the sum of stem-length and total length of all branches as well as by the quotient of total sapling leaf area over total shoot-length (i.e., leaf display efficiency). Specific leaf area (SLA) was calculated by dividing leaf blade area by blade mass. Relative water content of leaves was calculated by dividing the difference between fresh and dry leaf mass by the fresh leaf mass. Optical index values were converted to leaf CHL (Richardson et al., 2002). The difference of maximum CHL minus minimum CHL divided by the maximum was calculated as an index of phenotypic (or intra-crown) PPCHL and developmental (or organogenetic) DPCHL, respectively (Valladares et al., 2000, 2002; Bouman, 2014). The PPCHL within an individual sapling crown was calculated repeatedly by selecting, regardless of crown position, the minimum and maximum from among the four crown positions of a single (i.e., monthly or annually) data collection event. In the same way, the CHL values for September 2015 and 2016 were used to assess the effect of contrasting weather conditions (i.e., dry vs. wet year) on PPCHL. Four separate indices of DPCHL during the 2015 season were calculated for each crown position of each individual sapling by selecting minimum and maximum CHL from among the repeated measurements on the same leaf during its development May to September 2015. A DPCHL index was also calculated at the level of the whole sapling crown by selecting, regardless of crown position, minimum CHL and maximum CHL from among the repeated measurements on the same sapling. The difference between the average of the four leaf-level DPCHL indices minus crown-level DPCHL was divided by crown-level DPCHL as a crude estimate of the proportion of DPCHL that can be attributed to varying environmental exposure of presumably genetically identical leaves in the same sapling crown (cf., Klug and Cummings, 2002). Data were organized in MS Excel to compute descriptive statistics and produce figures. The CoStat software package (Cohort™ Software, Berkeley, CA; shareware) was used to perform analyses of variance (ANOVA) and calculate least significant differences (LSD) for separating means at a probability level of $p < 0.05$. Data were analyzed using a split-plot ANOVA design (Little and Hills, 1978) to determine effects of sapling treatment (i.e., AP-CON, AP-NPK, FG-CON, and FG-NPK; main plots) and sampling date (i.e., month or year; sub plots). Regression analysis was used to quantify relationships between selected plant traits. For this, the content of N and chlorophyll in the 4-leaf-sample collected from four crown positions was scaled up to sapling total foliar chlorophyll mass and total foliar N mass by accounting for the number of leaves per sapling. Thus, foliar chlorophyll mass and foliar N mass of each 4-leaf sample was multiplied by the number of leaves per sapling and then divided by four.

Table 1

Mean values of shoot characteristics for saplings of *Acer pensylvanicum* (AP) and *Fagus grandifolia* (FG) beneath mature hardwood canopy in East Bay Community Forest, Nova Scotia (Canada), June 2015; values in parentheses denote SE of mean ($n = 6$); p -values in bold indicate significance at $p = 0.05$.

	AP	FG	p-value
Basal area [mm ²]	129 (20)	202 (26)	< 0.001
Stem length (S) [m]	1.95 (0.20)	1.91 (0.17)	0.763
Rel. stem extension			
2015	0.06 (0.01)	0.08 (0.01)	0.409
2016	0.05 (0.01)	0.07 (0.01)	0.303
Total shoot length (T) [m]	4.07 (0.7)	11.87 (1.82)	< 0.001
S/T	0.62 (0.04)	0.20 (0.02)	< 0.001
No. of leaves	44 (8)	256 (31)	< 0.001
Leaf area [LA; m ²]	0.55 (0.09)	0.92 (0.14)	0.013
LA:T [cm ² cm ⁻¹]	12.7 (1.1)	8.2 (0.7)	0.001

3. Results

3.1. Canopy closure and sapling structure

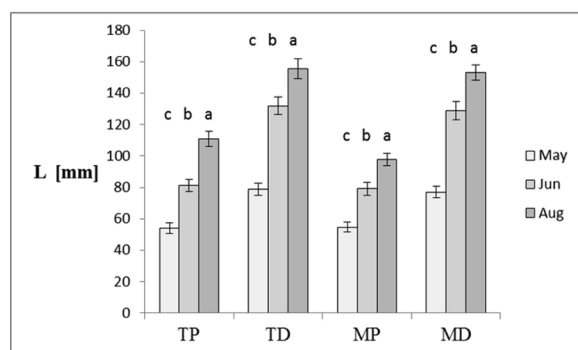
Quantitative estimates of forest canopy closure above each of the labelled saplings confirmed that saplings of all four treatments (i.e., AP-CON, AP-NPK, FG-CON, and FG-NPK) were situated in deep shade. The ANOVA showed no significant difference for means of forest canopy closure above sapling treatments ($p = 0.845$) although mean canopy closure showed a small but significant decrease from July to September ($p = 0.005$). Mean canopy closure ($n = 6$) ranged from 89.9 % above FG-NPK to 93.1 % above AP-NPK ($LSD_{0.05} = 8$ %) and decreased from 98 % in July to 84 % in September ($LSD_{0.05} = 9$ %). The shoot system of FG is carried by a significantly larger stem basal area than in AP (Table 1). However, saplings of AP and FG with comparable stem lengths did not differ significantly in their relative stem extension growth (i.e., leader growth) during the experimental period (i.e., 2015 and 2016). The inherent morphological divergence between the light-seeking and shade-adapted shoot systems becomes evident in the total length of the shoot system that is nearly three times as long in FG compared to AP. In fact, the mean quotient of stem length over total shoot length (i.e., S/T) was 0.20 (SE = 0.02) in FG compared to 0.62 (SE = 0.04) in AP hence AP invested approximately three times more resources in the development of stem tissues relative to the entire shoot system. The relationship between total shoot length and stem length was best described in AP by the linear regression equation $y = 3.42x - 2.58$ ($R^2 = 0.86$; $p < 0.05$) and in FG by $y = 10.85x - 8.05$ ($R^2 = 0.88$; $p < 0.05$). The striking difference between shoot morphologies coincided with markedly different patterns of foliation. Much more ramified FG crowns carry many more leaves than in AP although the difference in total sapling leaf area is less marked due to large leaf size in AP (Table 2). Total sapling leaf area scaled positively with total shoot length best described by the regression $y = 13.9x$ ($R^2 = 0.85$; $p < 0.05$) in AP and $y = 7.2x$ ($R^2 = 0.69$; $p < 0.05$) in FG hence AP displayed leaf area nearly twice as efficiently. Accordingly, the average leaf display efficiency (LA:T) differed significantly between FG (8.2; SE = 0.7) and AP (12.7; SE = 1.1) (Table 1). The two species also differed in their leaf phenology indicated by the seasonal development of leaf length (Fig. 2a and b). Leaf extension was completed earlier in FG than in AP ($p < 0.01$). Leaf length was a good predictor of leaf blade area best described by $y = 0.0024x^{2.215}$ ($R^2 = 0.91$) in AP and $y = 0.003x^{2.043}$ ($R^2 = 0.94$) in FG, respectively. The ANOVA showed a significant increase of leaf size ($p < 0.001$) with a concomitant decrease of SLA ($p < 0.001$) from proximal to distal crown positions in AP and FG (Table 2). Moreover, AP leaves contained significantly more water than FG. There was no significant crown position effect on RWC noted in the two species ($p = 0.771$). The mean water volume contained in total sapling foliage at apparent turgor was 24.5 cm³ (SE = 4.5) in AP compared to 21 cm³ (SE = 3.7) in FG. Application

Table 2

Mean values of physical leaf traits in four crown positions (MP = mid-proximal; TP = top-proximal; MD = mid-distal; TD = top-distal) for saplings of *Acer pensylvanicum* (AP) and *Fagus grandifolia* (FG) beneath mature hardwood canopy in East Bay Community Forest, Nova Scotia (Canada), August 2015; values in parentheses denote SE of mean ($n = 6$); italicized numbers indicate p -value of paired t -test; p -values in bold indicate significance at $p = 0.05$.

Position	AP	FG	AP vs. FG
Leaf size [cm^2]			
MP	65 (7)	18 (2)	< 0.001
TP	81 (9)	18 (1)	< 0.001
MD	159 (14)	53 (5)	< 0.001
TD	178 (15)	55 (5)	< 0.001
MP vs. TD	< 0.001	< 0.001	
Specific leaf area [$\text{m}^2 \text{ kg}^{-1}$]			
MP	44 (1)	43 (2)	0.519
TP	43 (2)	39 (2)	0.028
MD	40 (1)	37 (2)	0.088
TD	40 (2)	34 (1)	0.002
MP vs. TD	0.025	0.001	
Relative water content [g g^{-1}]			
MP	0.67 (0.00)	0.44 (0.02)	< 0.001
TP	0.67 (0.01)	0.42 (0.02)	< 0.001
MD	0.67 (0.00)	0.44 (0.01)	< 0.001
TD	0.68 (0.01)	0.42 (0.02)	< 0.001
MP vs. TD	0.185	0.606	

a



b

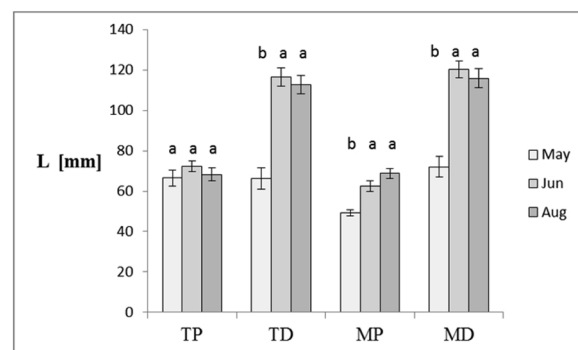


Fig. 2. Leaf length (L) measured on same twig in May, June, and August 2015 in mid-proximal (MP), top-proximal (TP), mid-distal (MD), and top-distal (TD) crown position of saplings of *Acer pensylvanicum* (AP; 2a) and *Fagus grandifolia* (FG; 2b) beneath northern hardwood canopy, East Bay Community Forest, Nova Scotia (Canada); vertical bars denote SE of mean ($n = 6$); same letter above columns of same crown position denotes mean monthly differences are non-significant at $p < 0.05$.

Table 3

Results of ANOVA on leaf chlorophyll concentration (CHL) for comparison of *Acer pensylvanicum* (AP) at natural soil fertility (AP-CON), AP with NPK application (AP-NPK), *Fagus grandifolia* (FG) at natural soil fertility (FG-CON), and FG with NPK application (FG-NPK) in mid-proximal (MP), top-proximal (TP), mid-distal (MD), and top-distal (TD) crown. Note: DF = degree of freedom; italicized values denote p -values; p -values in bold indicate significance at $p = 0.05$.

a) across months (May, June, July, August, and September 2015)				
Factor (DF)	MP	TP	MD	TD
Treatment (T; 3)	0.005	< 0.001	0.004	< 0.001
Month (M; 4)	< 0.001	< 0.001	< 0.001	< 0.001
M x T	< 0.001	< 0.001	< 0.001	< 0.001
b) across years (September 2015 and September 2016)				
Factor (DF)	MP	TP	MD	TD
Treatment (T; 3)	< 0.001	< 0.001	< 0.001	< 0.001
Year (Y; 1)	0.002	0.005	0.167	0.958
Y x T	0.626	0.427	0.016	0.009

of NPK exerted no significant effect on leaf size and RWC (data not shown). At $\text{LSD}_{0.05} = 55 \text{ cm}^2 \text{ g}^{-1}$, mean SLA of $443 \text{ cm}^2 \text{ g}^{-1}$ for AP-NPK was significantly different from AP-CON, FG-CON, and FG-NPK with mean values of 396, 387, $377 \text{ cm}^2 \text{ g}^{-1}$, respectively.

3.2. Optically assessed CHL

The ANOVA revealed significant effects of treatment and month on CHL in each crown position (Table 3a; Fig. 3a–d). There was a stronger increase for FG saplings compared to AP. In all four crown positions the seasonal CHL profile was most dynamic from 30 May to 4 August and remained flat until 19 September. The crown position effect was most marked in FG with an increase of mean CHL for top-distal leaves from $108 \text{ (SE = 5) mg m}^{-2}$ late May to $306 \text{ (SE = 14) mg m}^{-2}$ early August and for mid-proximal leaves from $94 \text{ (SE = 3) mg m}^{-2}$ late May to $267 \text{ (SE = 12) mg m}^{-2}$. Thus, there was approximately a 3-fold CHL increase regardless of crown position. Saplings receiving three split applications of NPK in 2015 showed no significant CHL response in AP (Fig. 3). On average, application of NPK elevated foliar CHL in mid-proximal crown position of FG (Fig. 3a) although the increase was statistically non-significant ($\text{LSD}_{0.05} = 33 \text{ mg m}^{-2}$). Comparing CHL across years confirmed the significant treatment effect. The effect of sampling year was significant for proximal leaves but this was not the case for distal leaves (Table 3b; data not shown). Thus, CHL values recorded for a new set of distal leaves on the same branch in 2016 were comparable to 2015.

3.3. Plasticity of optically assessed CHL

The ANOVA on monthly indices of PPCHL revealed significant treatment effects (Fig. 4; Table 4). Mean PPCHL indices varied considerably across months although the index was consistently higher in FG-CON compared to FG-NPK ($p = 0.010$). Thus, application of NPK reduced PPCHL in FG. For reference, mean intra-crown plasticity index values for SLA determined on leaves sampled in August 2015 were 0.23 ($\text{SE} = 0.02$) and 0.26 ($\text{SE} = 0.02$) for AP and FG, respectively. Morphometric leaf plasticity was well within the PPCHL range. There was no significant interaction between time and treatment hence experimental effects on treatments were consistent across months. The annual comparison of the PPCHL index between Sep. 2015 and Sep. 2016 confirmed the negative effect of NPK application on PPCHL in FG (Fig. 5). Most notably, there was no significant year effect albeit contrasting weather conditions (i.e., dry vs. wet year; Fig. 1). The ANOVA revealed a significant species effect on DPCHL (i.e., organogenetic

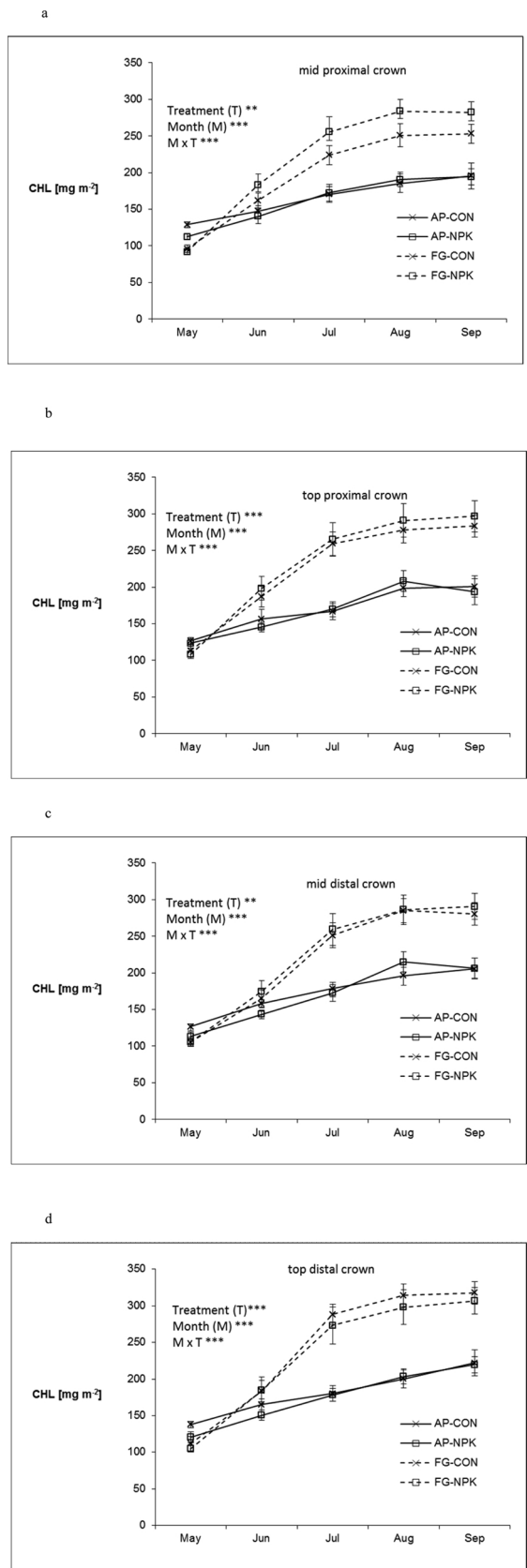


Fig. 3. Seasonal development of optically assessed foliar chlorophyll (CHL) content measured repeatedly on same leaf in 4 crown positions (mid proximal: a; top proximal: b; mid distal: c; top distal: d) of saplings of *Acer pensylvanicum* (AP) and *Fagus grandifolia* (FG) without fertilizer application (CON) and with fertilizer (NPK) beneath northern hardwood canopy, East Bay Community Forest, Nova Scotia (Canada), 2015; vertical bars denote SE of mean ($n = 6$).

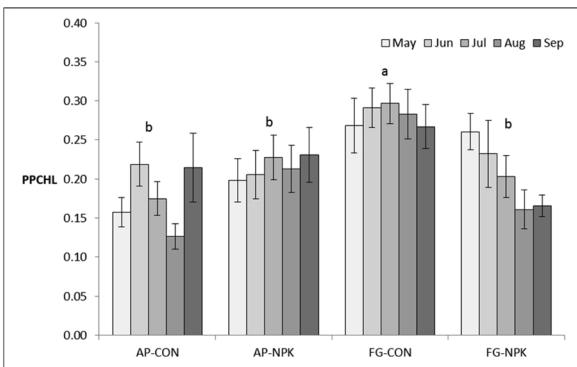


Fig. 4. Treatment effect on mean phenotypic plasticity index of leaf chlorophyll (PPCHL = (max – min)/max) by month in saplings of deciduous *Acer pensylvanicum* (AP) and *Fagus grandifolia* (FG) without fertilizer application (CON) and with fertilizer (NPK) in East Bay Community Forest, Nova Scotia, Canada, 2015; vertical bars indicate standard error of mean value; $n = 6$; same letter above columns denotes mean treatment differences are non-significant at $p < 0.05$.

Table 4
Results of ANOVA on phenotypic chlorophyll plasticity (PPCHL) within individual sapling crown for comparison of AP-CON, AP-NPK, FG-CON, and FG-NPK by month (May, June, July, August, and September 2015) and year (September 2015 vs. September 2016). P-values in bold indicate significance at $p = 0.05$.

Factor	Month	Year
Treatment (T)	0.010	0.151
Time (M/Y)	0.375	0.782
M/Y x T	0.358	0.930

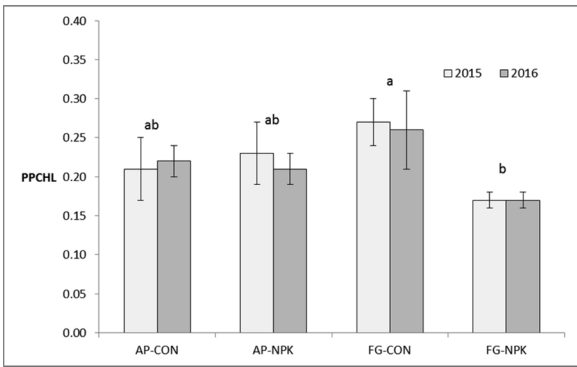


Fig. 5. Treatment effect on mean phenotypic plasticity index of chlorophyll (PPCHL = (max – min)/max) by year (Sep. 2015 vs. Sep. 2016) in saplings of deciduous *Acer pensylvanicum* (AP) and *Fagus grandifolia* (FG) without fertilizer application (CON) and with fertilizer (NPK) in East Bay Community Forest, Nova Scotia, Canada; vertical bars indicate standard error of mean value; $n = 6$; same letter denotes mean treatment differences are non-significant at $p < 0.05$.

PCHL) regardless of crown position ($p < 0.001$; Fig. 6). Separate estimates of DPCHL for leaves in four different crown positions varied within narrow ranges at an average leaf-level DPCHL of 0.43 and 0.64 in AP and FG, respectively. At the level of the whole sapling crown, mean DPCHL was 0.52 in AP compared to 0.72 in FG, respectively.

3.4. Foliar N

On a mass basis, ANOVA showed a significant treatment effect on N concentration ($p < 0.001$) with means of 19.6, 20.6, 13.0, and 15.3 mg N per gram for FG-CON, FG-NPK, AP-CON, and AP-NPK,

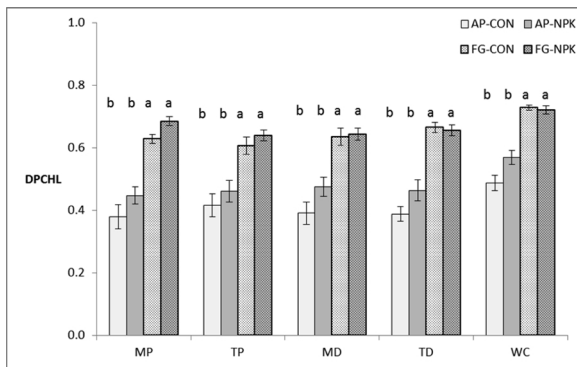


Fig. 6. Treatment effect on mean developmental plasticity index of leaf chlorophyll (DPCHL = (max – min)/max)) in four crown positions (MP = mid-proximal, TP = top-proximal, MD = mid-distal, and TD = top-distal) and whole crown (WC) of deciduous *Acer pensylvanicum* (AP) and *Fagus grandifolia* (FG) without fertilizer application (CON) and with fertilizer (NPK) at sapling stage during 2015 growing season (May to September) in East Bay Community Forest, Nova Scotia (Canada); vertical bars indicate standard error of mean value; n = 6; same letter above columns of same crown position denotes mean treatment differences are non-significant at $p < 0.05$.

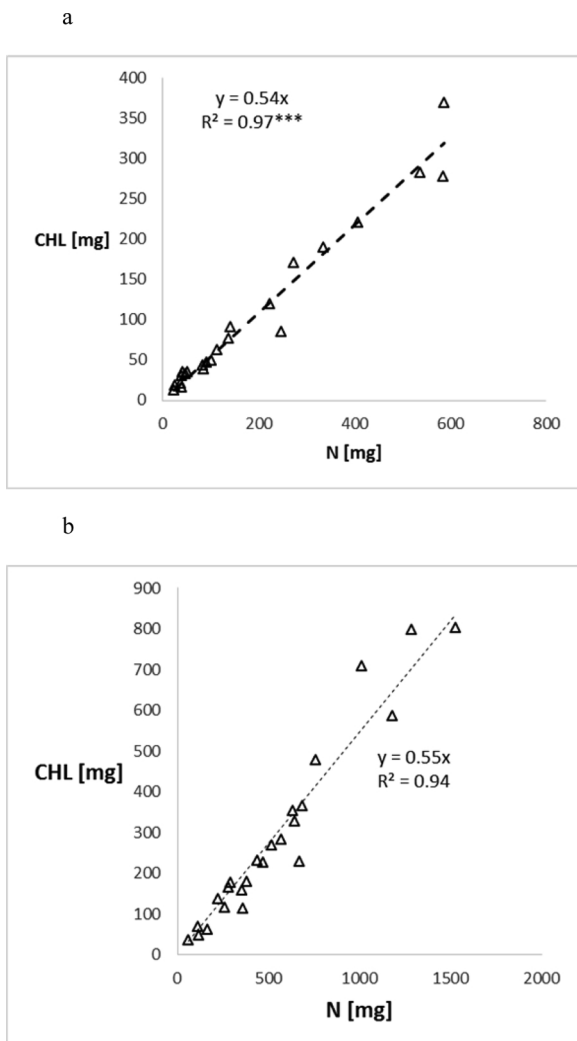


Fig. 7. Relationships between sapling total optically assessed chlorophyll mass and total foliar N mass for *Acer pensylvanicum* (7a) and *Fagus grandifolia* (7b) beneath northern hardwood canopy in East Bay Community Forest, Nova Scotia (Canada), August 2015; note scaling of axes differs between Fig. 7a and b.

respectively. An $LSD_{0.05}$ of 2.0 mg N per gram separated the species means and indicated a positive NPK effect on AP. Accounting for sapling crown foliage (i.e., number of leaves) in linear regressions on total sapling leaf chlorophyll mass over total sapling leaf N mass resulted in very similar slope values of 0.54 for AP (Fig. 7a) and 0.55 for FG (b). The species effect was also significant for foliar N content per unit leaf area ($p < 0.001$) with mean values of 355, 362, 548, and 572 mg N per m^2 for AP-CON, AP-NPK, FG-CON, and FG-NPK, respectively. An $LSD_{0.05} = 53 \text{ mg m}^{-2}$ did not confirm a significant NPK effect on AP.

4. Discussion

4.1. Morphometric adaptations to deep shade

The morphological characterization of sapling crown and foliage for AP and FG is in good agreement with previous research in Quebec (Lei and Lechowicz, 1990; Beaudet and Messier, 1998) and Virginia (King, 2003). High leaf display efficiency (i.e., LA:T; Table 1) for AP saplings comes close to values reported by Lei and Lechowicz (1990) for the same species in the understory beneath a northern hardwood canopy dominated by *A. saccharum* and FG. Moreover, low stem extension values for AP (Table 1) align well with values reported by King (2003) for AP saplings growing in deep shade. In the same vein, the significant increase of AP SLA from distal to proximal crown position is consistent with decreasing LMA (the inverse of SLA) from gap centre to gap edge in Lei and Lechowicz (1998) although the crown position effect in this study is not as strong as the canopy effect reported by Lei and Lechowicz (1998). Low leaf display efficiency in FG saplings concurs with results reported by Beaudet and Messier (1998) for FG saplings in the understory of a sugar maple–yellow birch–beech stand in Quebec. The intra-crown differentiation of SLA in this study also falls within the value range reported by Beaudet and Messier (1998) for FG at a light exposure of less than 10 % of the photosynthetic photon flux density (PPFD) above a broad-leaved forest canopy. Non-significant variation of stem extension at low rates between AP and FG (Table 1) is also indicative of an overall low light regime and limited spatial heterogeneity of growing conditions among saplings inferred from forest canopy closure. The forest had not experienced forest canopy disturbances for more than 15 years and gap border trees closed in on existing gaps used in a previous experiment within the same forest (Bouman et al., 2012). Beaudet et al. (2004) did not note any significant variation in light profiles 13 years following small canopy openings due to selective tree harvest in a northern hardwood.

4.2. Shade tolerance

The CHL difference between AP and FG (Fig. 3) corroborates the results of a physiological experiment reporting significantly lower photosynthetic rates in AP than in FG (Jurik et al., 1988). A concordance of CHL and photosynthetic rates is consistent with previous research on *F. sylvatica* (Epron et al., 1996). Higher late-season CHL in FG compared to AP was consistent with a late-season comparison of foliar pigmentation for the same species pair in Massachusetts by Lee et al. (2003) although the authors did not monitor CHL from early to late season. The observed species differences in seasonal leaf length extension (Fig. 2) and concomitant seasonal leaf CHL filling (Fig. 3) support the conclusion by Lopez et al. (2008) that leaf phenology contributes to shade tolerance. The authors found that early leafing greatly increased interception of growing season photon flux before canopy closure by temperate deciduous tree species in a Southern Appalachian forest. Thus, FG-specific leaf phenology (i.e., rapid leaf expansion; Fig. 2) allows for light-induced chlorophyll filling at a high rate prior to full forest canopy closure in early summer (Fig. 3) which in turn induces high seasonal DPCHL in FG (Fig. 6). High DPCHL within a sapling crown displaying a large leaf area (Table 1) renders FG saplings

more tolerant of shade and distinguishes the species from the transient AP with its late leaf phenology, low DPCHL, and light-seeking crown morphology. Late season values for PPCHL and phenotypic SLA plasticity in FG (Fig. 5, Table 2) were in good agreement with values reported for *F. sylvatica* (Jarčuška, 2011; Bouman, 2014). High DPCHL in FG saplings (Fig. 6) is consistent with observations by Valladares et al. (2002) during an acclimatisation experiment on *F. sylvatica* and corroborates the authors' assertion that high chlorophyll plasticity enhances shade tolerance. Nearly constant DPCHL across the four crown positions verified the assumption of genetic homogeneity within sapling crowns. Interestingly, the difference between the average leaf-level DPCHL and the crown-level DPCHL shown in Fig. 6 is very similar for AP and FG. Hence, the relative environmental proportion on DPCHL was 18 % in AP and 11 % in FG. Conversely, strong genetic control of DPCHL appeared higher in FG than in AP.

The close relationship in the regression of sapling chlorophyll mass over N mass in AP and FG (Fig. 7a and b) corroborates previous research on *F. sylvatica* (Bouman, 2014). The relationship is consistent with the general notion of a close association of CHL and proteins at the cellular level. However, NPK applications on PPCHL played out differently between the two species. Mean CHL in FG responded positively to NPK applications most markedly in the most light-limited crown position (i.e., MP; Fig. 3a) which in turn reduced the intra-crown plasticity index significantly compared to the FG control (Fig. 4). Hence, co-limitation of plant growth by mineral nutrients was still a factor in deep shade for FG likely due to a combination of morphometric and physiological trait acclimatisation to shade beyond the scope of this study. Significantly lower RWC in FG compared to AP might have anatomical implications for light harvesting capacity and photosynthetic potentials in addition to species differences in leaf mass fraction on whole sapling biomass, light compensation point etc. (Niinemets and Sack, 2006). However, mineral nutrition was non-limiting in deep shade for AP (Körner, 2008) thereby corroborating a lower shade tolerance rank for AP than for FG (Niinemets and Valladares, 2006). High N content in large sapling foliage and an elaborated shoot system necessitate N conservation hence selecting for low plasticity of growth rate but high PCHL in FG very similar to its congeneric *F. sylvatica* (Valladares et al., 2002; Bouman, 2014). Apparently, water availability was not a site factor differentiating PPCHL. Re-measurement of the same saplings in the dry year 2016 showed no significant year effect on PPCHL compared to the wet year 2015. Moreover, the inverse differences for total leaf area and leaf RWC resulted in similar total foliage water volumes per sapling for AP and FG.

5. Conclusion

Results support the hypothesis of higher plasticity of leaf chlorophyll (PCHL) in *Fagus grandifolia* (FG) than in *Acer pensylvanicum* (AP). A markedly more dynamic CHL increase coincided with more rapid leaf expansion in FG compared to AP. Differential leaf phenology likely contributed to high PCHL in FG. Results also support, in the case of FG, the hypothesis that nutrient-poor soil conditions favour phenotypic PCHL. This is however, not true for AP and confirms a lower shade tolerance rank for AP. Results strongly suggest that the ecological conditions for advanced regeneration in deep shade on nutrient-poor soils maintain high PCHL in FG populations.

Author's contribution

Thomas Bouman is the sole author of the manuscript who designed the experiment, collected the data, analyzed the data, and wrote the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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